

# California Housing Price Prediction Using Linear Regression

## Internship Project Report

Artificial Intelligence & Machine Learning - Task 1

|                     |                                                |
|---------------------|------------------------------------------------|
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| <b>Project Type</b> | Machine Learning Regression Project            |
| <b>Dataset</b>      | California Housing Dataset                     |
| <b>Algorithm</b>    | Linear Regression                              |
| <b>Tools</b>        | Python, Jupyter Notebook, Pandas, Scikit-Learn |

## Abstract

This project focuses on developing a machine learning based house price prediction system using the California Housing Dataset. The project follows a complete machine learning workflow including data loading, exploratory data analysis, preprocessing, model training, evaluation, visualization, and report generation. Linear Regression is used as the baseline regression model to predict median house values.

## Objective

The main objective of this project is to build and evaluate a Linear Regression model that can predict median house values using district-level demographic and geographic features from California.

## Problem Statement

House price prediction is a real-world regression problem where the goal is to estimate a continuous numerical value. In this project, the target variable is median house value, and the input features include income, house age, average rooms, population, occupancy, latitude, and longitude.

## Dataset Feature Table

| Feature     | Description                              |
|-------------|------------------------------------------|
| MedInc      | Median income in block group             |
| HouseAge    | Median house age in block group          |
| AveRooms    | Average number of rooms per household    |
| AveBedrms   | Average number of bedrooms per household |
| Population  | Block group population                   |
| AveOccup    | Average number of household members      |
| Latitude    | Geographic latitude                      |
| Longitude   | Geographic longitude                     |
| MedHouseVal | Target variable: median house value      |

## Dataset Description

The California Housing Dataset contains 20,640 records and 8 numerical features. The target variable is MedHouseVal, which represents median house value in units of 100,000 dollars. The dataset is available directly through the scikit-learn library using `fetch_california_housing()`.

## Methodology

The project methodology includes importing required libraries, loading the dataset, converting it into a pandas DataFrame, performing data inspection, checking missing values, conducting exploratory data analysis, splitting the dataset into training and testing sets, training the Linear Regression model, evaluating the model using MAE, RMSE, and R2 Score, visualizing predictions, and saving the trained model.

## Exploratory Data Analysis

Exploratory Data Analysis is performed using summary statistics, distribution plots, scatter plots, and a correlation heatmap. The analysis helps identify feature relationships and important trends. Median Income generally shows the strongest positive relationship with house value.

## Model Development

The dataset is split into 80 percent training data and 20 percent testing data. A Linear Regression model is trained on the training set and evaluated on the unseen test set. Linear Regression is selected because it is simple, interpretable, and suitable for predicting continuous values.

## Evaluation Metrics

The model is evaluated using Mean Absolute Error, Root Mean Squared Error, and R2 Score. MAE measures average absolute prediction error, RMSE penalizes larger errors, and R2 Score shows how much variation in the target variable is explained by the model.

## Results Template

After running the notebook, copy the final metrics here. Typical baseline results for this dataset are around MAE = 0.53, RMSE = 0.75, and R2 Score = 0.58 to 0.61. Your exact values may vary slightly depending on environment and split settings.

## Conclusion

The Linear Regression model successfully demonstrates a complete machine learning workflow for house price prediction. The model provides moderate predictive performance and works well as a baseline model. The analysis shows that median income and location are important factors affecting house prices.

## Future Scope

The project can be improved using feature engineering, scaling, regularized regression, Random Forest Regression, Gradient Boosting, XGBoost, hyperparameter tuning, and deployment through a web application such as Streamlit or Flask.

## References

Scikit-Learn California Housing Dataset Documentation; Scikit-Learn Linear Regression Documentation; Pandas Documentation; NumPy Documentation; Matplotlib and Seaborn Documentation.